Examiner reports

Q1.

Part (a) was completed very successfully, with around three quarters of students scoring full marks. The most common mistakes seen were using formulae for geometric sequences or the formula for U_n rather than S_n. Those who used the correct formulae were generally able to set up the correct equation and simplify correctly.

Part (b) discriminated well between students of differing abilities. Most who had been at least partially successful in part (a) made good progress in forming a second equation from the extra information. Elimination of either a or d from their simultaneous equations initially resulted in an unwieldy equation. This gave a perfect opportunity for students to make efficient use of technology and avoid several lines of routine manipulation. Unfortunately, many students chose to manipulate rather than solve directly, often resulting in errors. Students should feel confident that they will not be penalised for solving a quadratic equation on their calculators, unless full justification is explicitly requested.

Q3.

Students almost always scored full marks for their answers to parts (a)(i) and (a)(ii) but writing the nth term of the geometric series in the required form in (a)(iii) proved to be much more challenging. Although the majority of students started correctly by writing 420 × 0.7^{n-1}, a large proportion of these students failed to write this in the form 600 × 0.7^n. The most common wrong answer, for those who attempted to re-write 420 × 0.7^{n-1} in the stated form, was 294 × 0.7^n. In part (b)(ii) there was some confusion with the term and sigma notation which resulted in the sum being equated to zero. The most successful approach was to solve U_k = 0 to obtain the value of the positive integer k and then to find the value of $\frac{k}{2}$ (240 + 0), which represents $\sum_{n=1}^{k} u_n$, the sum of the k terms of the arithmetic series with 1st term 240 and last term 0(= U_k) . In a minority of cases where the correct expression for U_k was equated to zero, an incorrect rearrangement resulted in a negative value for k. Students who made this error did not seem to realise that such a value for k was meaningless.

Q4.

Part (a) was a good source of marks for most candidates, with the formula for the sum of an arithmetic series being well understood and the necessary stage(s) needed in proving a printed answer being appreciated. Part (b) was also well answered, though a number of candidates wrongly used u_5 as a + 5d or, worse still, used the formula for S_5, the sum to five terms. Not all candidates who obtained the two correct linear equations were able to correctly solve them simultaneously.

Part (c), as expected, proved to be the least well answered part of the question, with a number of candidates making no attempt to answer it. Those candidates who failed to replace $\sum_{n=1}^{25} u_n$ by 3500 rarely made any progress. The most common error was to replace $\sum_{n=1}^{25} u_n$ by 325, presumably obtained by applying the irrelevant formula $\sum_{r=1}^{n} r = \frac{1}{2}n(n+1)$ which appears in the formulae booklet.

Q5.

This question, which tested the use of standard formulae for arithmetic series, proved to be a welcome opener with a very large majority of students scoring full marks. As ever, those students who stated the relevant general formula before substituting the values put themselves at an advantage if they then went on to make a numerical slip. There were a few cases of students finding the sum to 100 terms in part (c), but this was only penalised by one mark. Most other losses of mark were due to arithmetical errors.

Q6.

It was pleasing to see a very high proportion of candidates showing sufficient detail in their solutions to completely justify their obtaining of the printed result in part (a). Most candidates used the correct general formula for the nth term of an arithmetic series to find u_{16} and u_{21} in part (b), but it was not uncommon to see the multiplier 2 being applied to u_{21} instead of u_{16}. The final part of the question was answered poorly with many candidates either solving $u_k = 0$ instead of $S_k = 0$, or looking in the formulae booklet and writing $\frac{k}{2}(k + 1) = 0$.

Q7.

Many candidates correctly showed the printed value for r in part (a)(i) by forming and solving the equation '$\frac{10}{1-r} = 50$'. Those who used verification methods rarely scored both marks. Most candidates correctly found the second term of the geometric series, but a significant number of candidates in part (b)(i) tried to 'guess' the answer for the common difference instead of using the formula for the n^{th} term of an arithmetic series and ended up with the wrong answer '$d = 0.5$'.

The final part of the question, which involved understanding of the sigma sign, was better answered than in previous papers but still the majority of candidates failed to obtain the correct value. The most common wrong answer was 820, obtained by using $\frac{1}{2}n(n + 1)$ with $n = 40$, presumably from scanning through the formulae booklet.

Q8.

In part (a), many candidates applied at least one correct formula, but it was not uncommon to see $S_n = \frac{n}{2}(a + l)$ quoted and used as though l, the last term, was 1, or l not replaced by $a + (n - 1)d$. A significant number of average candidates attempted a verification or just used '$38 = a + 24 \times 1.5$' to get '$a = 2$' and then applied the same general equation to get the circular argument '38 = 2 + 24d so $d = \frac{38-2}{24} = 1.5$'.

Some better candidates correctly obtained the two equations '38 = a + 24d' and '1250 = 20(2a + 39d)' but then just wrote down 'd =1.5, a = 2'. As 'd = 1.5' is printed in the question, candidates were required to show sufficient working for the solution of these simultaneous equations before the final two marks could be awarded. The most common error in part (b) was candidates linking S_n to 100 rather than using $u_n < 100$. However, it was pleasing to see many of those candidates who could not answer part (a) using the printed value for d to answer part (b).

Q9.

Parts (a) and (b) were answered very well with far fewer candidates than in previous years confusing the formulae for the nth term of an arithmetic series with that for the sum of the series to n terms. In part (c), those candidates who considered the sum of the last 100 terms as an arithmetic series, with first term 751 and last term 1444, were generally more successful than those candidates who considered it as $S_{200} - S_{100}$. In the latter case, the correct value, 149500, for S_{200} was usually found, but the common error in finding S_{100} was to take the last term as 1444 rather than 744.

Q10.

Those candidates who used the formula $S_n = \frac{1}{2}n[2a + (n-1)d]$ generally went on to show sufficient detail to reach the printed answer in part (a). Part (b) was answered less well with a significant minority interpreting the sum of the second term and the seventh term is 13' as $S_2 + S_7 = 13$ instead of $u_2 + u_7 = 13$. It was also quite common to find candidates correctly finding the equation $2a + 7d = 13$ but then not realising that this equation needed to be solved simultaneously with the given equation in part (a) to find the value of a and the value of d. A few candidates clearly misread 'seventeenth' for seventh' in the question but the resulting penalty was not substantial.

Q11.

Part (a) was answered well by either counting on or by using the formula for the nth term, u_n, of an arithmetic series as given on page 4 of the formulae booklet. However, it was surprising to see a significant number of candidates 'unable' to evaluate 1 + (9)6 correctly. Although some candidates used S_{V}) instead of u_{10}, this type of error was less common than in previous years. Part (b)(i) was answered well by the average and better candidates but many answers to part (b)(ii) consisted of non-integer values of n or more than one value of n.

Q12.

Part (a) was generally well answered although some tried to use formulae for geometric series rather than arithmetic series. Most were able to find the sum to 20 terms in part (b) although arithmetical errors were noted. Part (c), which involved sigma notation, was not always attempted. The common errors seen otherwise were subtracting S_{21} rather than S_{20} and, normally only produced by the weaker candidates, using the formula for the nth term instead of the sum to n terms. Some candidates wasted time by showing that the sum to 50 terms was indeed the given value, 11525.

Q13.

Part (a) was answered extremely well, but the wrong answer, 3, for the common difference was given more often than the correct answer. Part (c) was answered badly. The most common wrong method was to ignore the sigma sign and just solve $90 - 3n = 0$. There were, however, some very elegant solutions seen based on the symmetry of the sequence. Other candidates

started with a correct first step by using the formulae $S_n = \frac{n}{2}(a + 1)$ from the formulae booklet to reach $\frac{1}{2}k(87 + 90 - 3k) = 0$, but then multiplied out the brackets and abandoned the solution after a mass of algebra. Those using $S_n = \frac{n}{2}[2a + (n-1)d]$ often gained the equivalent method marks but ended up with a negative value for k after using the incorrect value d = 3.